

B. Alternative, prior, and acquisition conditions

Archived codes by workflow. Each task row has CANN and CUDA columns; G uses ROCm/HIP in the second column.

| Task                                           |                            | M5 Alternatives |       | M6 Credibility |       | M7 Prior* |       | M8 Effort |       |
|------------------------------------------------|----------------------------|-----------------|-------|----------------|-------|-----------|-------|-----------|-------|
|                                                |                            | CANN            | CUDA‡ | CANN           | CUDA‡ | CANN      | CUDA‡ | CANN      | CUDA‡ |
| Environment and installation                   |                            |                 |       |                |       |           |       |           |       |
| I                                              | Version compatibility      | 3               | 2     | 4              | 4     | 2         | 4     | 3         | 3     |
| J                                              | Installation               | 3               | 3     | 4              | 4     | 4         | 5     | 5         | 5     |
| K                                              | Container setup            | 2               | 3     | 2              | 4     | 3         | 5     | 1         | 4     |
| Operator development                           |                            |                 |       |                |       |           |       |           |       |
| C                                              | Library selection          | 3               | 5     | 4              | 4     | 3         | 5     | 5         | 5     |
| D                                              | Custom operator            | 3               | 4     | 2              | 4     | 2         | 5     | 1         | 5     |
| L                                              | Operator accuracy          | 3               | 2     | 4              | 4     | 3         | 4     | 4         | 1     |
| M                                              | Dynamic shapes / tiling    | 2               | 2     | 4              | 4     | 2         | 5     | 3         | 5     |
| N                                              | Operator fusion            | 2               | 2     | 3              | 4     | 2         | 4     | 5         | 1     |
| O                                              | Framework integration      | 3               | 2     | 5              | 4     | 3         | 5     | 1         | 3     |
| Training                                       |                            |                 |       |                |       |           |       |           |       |
| F                                              | Distributed training       | 3               | 3     | 3              | 4     | 3         | 4     | 5         | 5     |
| P                                              | Mixed precision            | 3               | 3     | 4              | 4     | 4         | 5     | 2         | 4     |
| Q                                              | Memory / OOM               | 3               | 3     | 4              | 4     | 3         | 5     | 3         | 2     |
| R                                              | Training convergence       | 3               | 3     | 4              | 3     | 3         | 5     | 3         | 4     |
| Inference and deployment                       |                            |                 |       |                |       |           |       |           |       |
| A                                              | Model conversion           | 4               | 4     | 4              | 4     | 3         | 4     | 4         | 5     |
| H                                              | Quantization               | 4               | 3     | 3              | 4     | 3         | 4     | 4         | 5     |
| S                                              | Inference serving          | 2               | 3     | 4              | 4     | 2         | 4     | 4         | 5     |
| T                                              | Dynamic inference          | 3               | 2     | 5              | 4     | 3         | 5     | 3         | 1     |
| Performance and optimization                   |                            |                 |       |                |       |           |       |           |       |
| B                                              | Profiling                  | 3               | 4     | 3              | 4     | 3         | 4     | 4         | 5     |
| U                                              | Memory / occupancy         | 3               | 3     | 4              | 4     | 3         | 5     | 4         | 5     |
| V                                              | Compute-transfer overlap   | 3               | 3     | 4              | 4     | 3         | 5     | 4         | 4     |
| W                                              | Multi-device communication | 3               | 3     | 4              | 4     | 3         | 4     | 3         | 5     |
| Debugging                                      |                            |                 |       |                |       |           |       |           |       |
| E                                              | Error-code diagnosis       | 3               | 4     | 3              | 4     | 2         | 4     | 4         | 5     |
| X                                              | Memory-error diagnosis     | 2               | 3     | 4              | 4     | 2         | 4     | 4         | 5     |
| Migration                                      |                            |                 |       |                |       |           |       |           |       |
| Y                                              | Chip migration             | 3               | 3     | 4              | 4     | 3         | 5     | 1         | 4     |
| Z                                              | Programming concepts       | 3               | 3     | 4              | 4     | 4         | 5     | 1         | 1     |
| Migration analogy (not in CANN/CUDA summaries) |                            |                 |       |                |       |           |       |           |       |
| G                                              | Migration analogy          | 3               | 3     | 4              | 4     | 3         | 4     | 5         | 5     |

M1–M10 are archived ordinal codes (1–5; higher is more favorable; M8 is an inverse effort score). "—" = unobserved core adequacy.

\* M7 is an archived estimate. † M11 is the historical heuristic availability index. ‡ For G, CUDA‡ = ROCm/HIP; it is excluded from CANN/CUDA summaries.